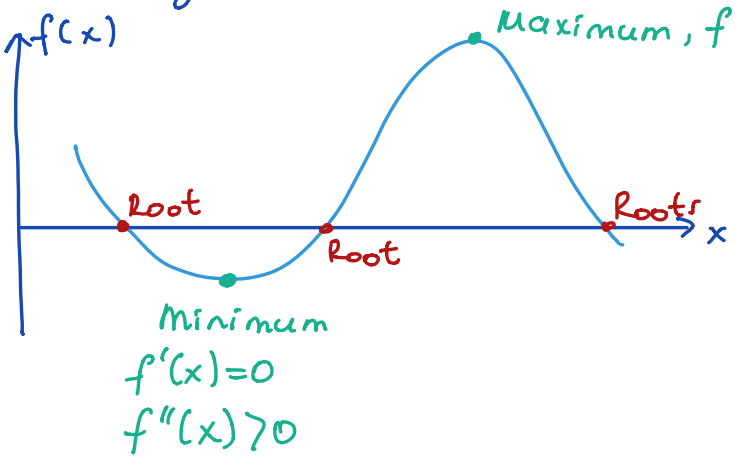
CHAPTER 7 – Optimization

- Mathematical perspective: optimization deals with finding the minima and maxima of a function that depends on one or more variables. The goal is to determine the values of the variables that give the minima or maxima of the function.



Relationship between roots and optima:

- Differentiate the function and locate the root (the zero)

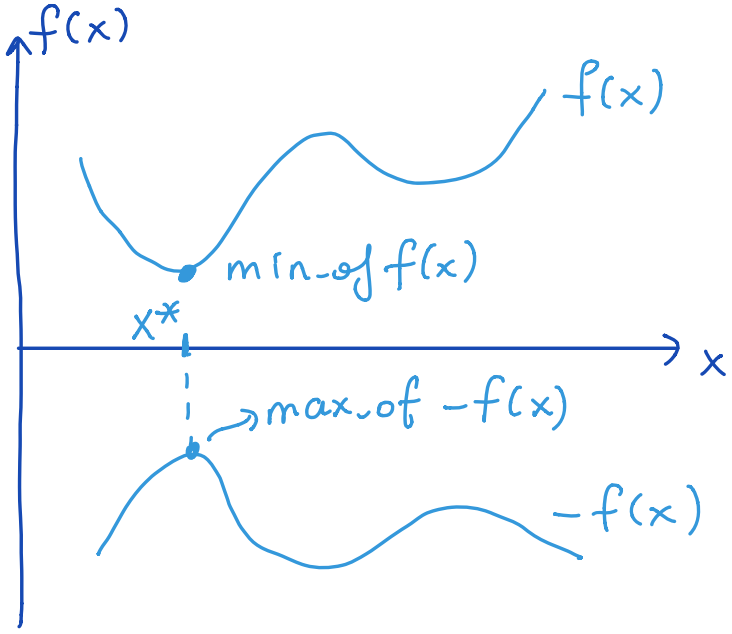
$f'(x)=0$ ($f'(x)$ is generally not available analytically)

- Direct optimization methods are available.

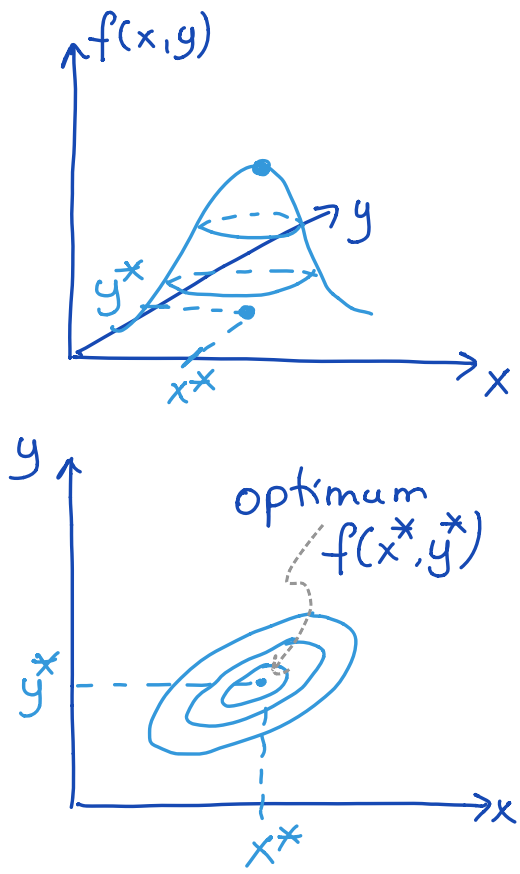
One-dimensional problems: involve functions that depend on a single dependent variable.

Multi-dimensional problems: involve functions that depend on two or more dependent variables.

One-dimensional optimization



2D Optimization



* Finding a maximum or minimum are identical. x^* minimizes $f(x)$ and maximizes $-f(x)$

Optimization – Mathematical Background

An optimization or mathematical programming problem can be stated as:

Find $\underline{x}$, which minimizes or maximizes $f(\underline{x})$ subject to:

$$d_i(\underline{x}) \leq a_i \quad , i = 1, 2, \ldots, m \qquad \underline{x} \in \mathbb{R}^n$$

$$e_i(\underline{x}) = b_i \quad , i = 1, 2, \ldots, p$$

$\underline{x}$: n dimensional vector

$f(\underline{x})$: objective function
$d_i(\underline{x})$: inequality constraints
$e_i(\underline{x})$: equality constraints
a_i, b_i : constants

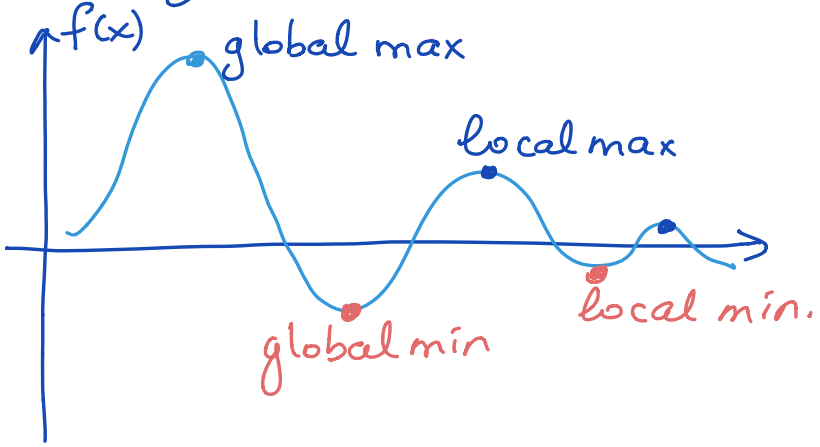
- If $f(\underline{x})$ and the constraints are linear, we have linear programming.
- If $f(\underline{x})$ is quadratic and the constraints are linear, we have quadratic programming.
- If $f(\underline{x})$ is not linear or quadratic and/or the constraints are nonlinear, we have nonlinear programming.
- If constraints are included, we have constrained optimization, otherwise it is an unconstrained optimization problem.
- Constrained problems: Degree of freedom = $n-p-m$

To obtain a solution $p+m \leq n$. If $p+m > n$, the problem is said to be overdetermined.

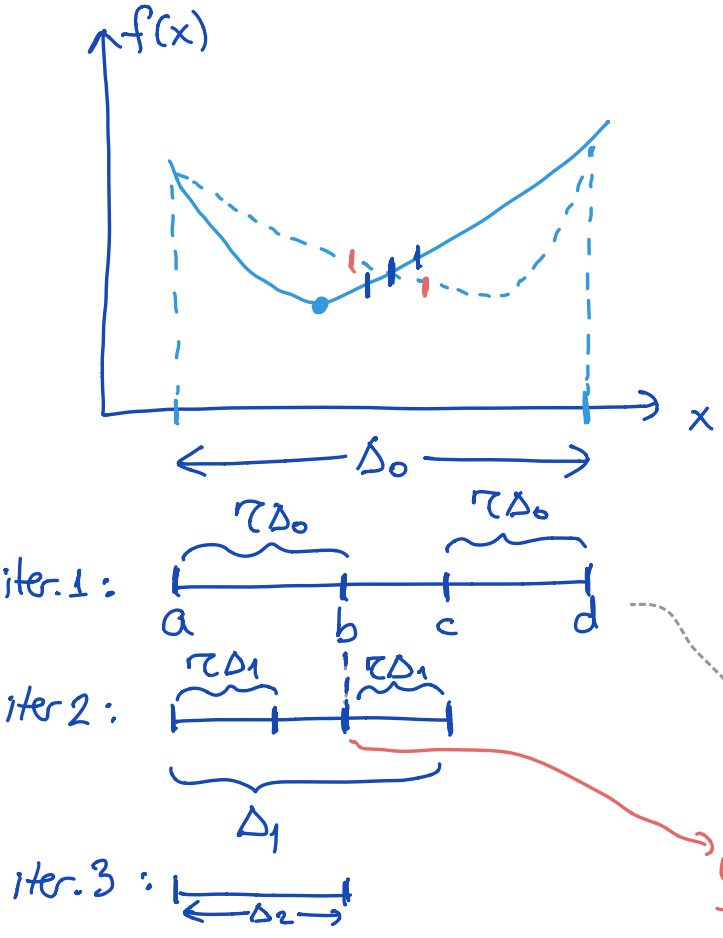
One-dimensional Optimization

Global optimum: the very best solution.
Local optimum: though not the very best, better than its immediate neighbors.

- we are interested in finding the global minimum.
- We do not want to mistake a local optimum for the global optimum.



Golden Section Method (Michael Zibulewsky)



- Evaluating a function at a single intermediate value will not tell whether the min. is in the left half or right half.
- Evaluate at two intermediate pts and get an idea of slope.

if $f(b) < f(c)$
choose ac
else
choose bd

We want to carry pt. b to the next iteration.

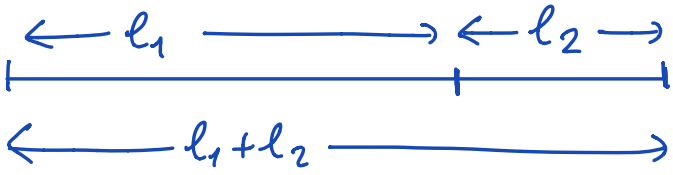
$$\Delta_1 = \Delta_0 - \tau\Delta_0 = (1-\tau)\Delta_0$$

$$\tau\Delta_0 = \Delta_1 - \tau\Delta_1 = (1-\tau)\Delta_1 = (1-\tau)(1-\tau)\Delta_0$$
$$= (1-\tau)^2\Delta_0$$

$$\tau\cancel{\Delta_0} = (1-\tau)^2\cancel{\Delta_0} \quad \Rightarrow \tau = (1-\tau)^2 \Rightarrow \tau = \frac{3-\sqrt{5}}{2} = 0.3820$$

Most efficient method to find the minimum, when you only have function evaluations (not derivative evaluations)

Golden-ratio:



Euclid's golden ratio: divide a line so that

$$\frac{l_1+l_2}{l_1} = \frac{l_1}{l_2}$$

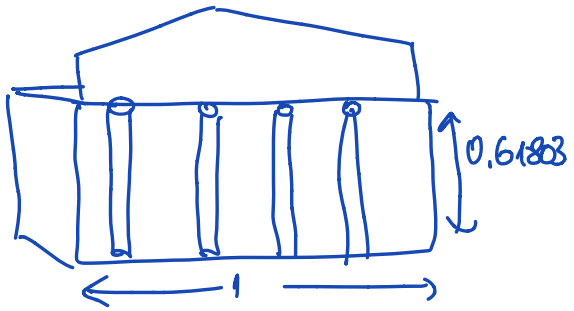
Multiply by $\frac{l_1}{l_2}$ and collect terms:

$$\underbrace{\frac{l_1}{l_2}}_{\phi} + 1 = \frac{l_1^2}{l_2^2}$$

$$\phi = \frac{l_1}{l_2}$$

$$\phi^2 - \phi - 1 = 0 \;, \Rightarrow \phi = \frac{1+\sqrt{5}}{2} = 1.618$$

Parthenon in Athens:



ϕ-1 : Golden ratio